

1. Write a program to create a class/structure `TreeNode` to represent a node in a binary tree. Each node should have three attributes: `data`, `left`, and `right`. Then, create a class/structure `BinaryTree` to represent the binary tree itself. Implement the following operations:

1. `insert(data)`: Insert a new node with the given data into the binary tree (assume a simple insertion without balancing).

2. `pre_order_traversal()`: Perform a pre-order traversal of the tree and print the data of each node.

Code:

```
main.c  [ ] [ ] [ ] Share Run
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  // 1. Structure for a Tree Node
5  struct TreeNode {
6      int data;
7      struct TreeNode* left;
8      struct TreeNode* right;
9  };
10
11 // Function to create a new node
12 struct TreeNode* createNode(int data) {
13     struct TreeNode* newNode = (struct TreeNode*)malloc(sizeof(struct
        TreeNode));
14     newNode->data = data;
15     newNode->left = NULL;
16     newNode->right = NULL;
17     return newNode;
18 }
19
20 // 2. Function to insert data (Simple BST insertion)
21 struct TreeNode* insert(struct TreeNode* root, int data) {
22     if (root == NULL) {
23         return createNode(data);
24     }
25 }
```

```
26     if (data < root->data) {
27         root->left = insert(root->left, data);
28     } else {
29         root->right = insert(root->right, data);
30     }
31
32     return root;
33 }
34
35 // 3. Pre-order Traversal (Root -> Left -> Right)
36 void preOrderTraversal(struct TreeNode* root) {
37     if (root == NULL) {
38         return;
39     }
40
41     printf("%d ", root->data); // Process Root
42     preOrderTraversal(root->left); // Process Left
43     preOrderTraversal(root->right); // Process Right
44 }
45
46 int main() {
47     struct TreeNode* root = NULL;
48
49     // Inserting data
50     root = insert(root, 10);
51     root = insert(root, 5);
```

```
51     insert(root, 5);
52     insert(root, 15);
53     insert(root, 3);
54     insert(root, 7);
55
56     printf("Pre-order Traversal: ");
57     preOrderTraversal(root);
58     printf("\n");
59
60     return 0;
61 }
```

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Output:

Output

Pre-order Traversal: 10 5 3 7 15

=== Code Execution Successful ===